Lecture 4 Scribe

L4.pdf
PDF

Generate a lecture scribe intended to serve as reference material for exam preparation using ONLY the provided context (Lecture 12 slides). Instructions: 1. Purpose: The scribe must be exam-ready, strictly following the content of the slides. 2. Content: Include definitions, assumptions, theorem statements, proofs or derivations, and worked examples exactly as presented in the lecture materials. 3. Constraints: Do not introduce new examples, outside information, or hallucinations. Do not use phrases like 'for example' to add your own content. 4. Structure: Organize the scribe clearly with appropriate headings corresponding to the lecture flow. Goal: Produce a clean, accurate revision document based solely on the attached file."

Below is a **clean, exam-ready lecture scribe** prepared strictly from the provided Lecture 12 slides.

CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability [L4]

Instructor: Dhaval Patel, PhD
Date: January 15, 2026 [L4]

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1. Course Context and Motivation

1.1 Why Learn CSE400?

- Daily life conversations involve uncertainty.
- Engineering applications include:
  - Speech recognition
  - Radar systems
  - Communication networks [L4]

1.2 Engineering Applications

Speech Recognition System

Templates of spoken words (e.g., hello, yes, no, bye) are compared across different speakers to identify patterns. [L4]

Radar System

Radar detects objects by transmitting signals and analyzing reflected waves. [L4]

Communication Network

Data travels from a source to a destination through multiple network nodes. L4

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2. Lecture Outline L4

Introduction to Probability Theory

- Experiments, sample space, and events
- Axioms of probability
- Corollaries and propositions
- Assigning probability: classical & relative frequency approaches

Joint Probability

- Motivation, notation, and concepts
- Example: Card deck
- Example: Costume party

Conditional Probability

- Motivation, notation, and concepts
- Cards without replacement
- Game of poker
- The missing key

(Note: Slides provided focus on the Introduction section.)

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3. Introduction to Probability Theory

3.1 Experiments, Outcomes, and Events

Definition: Experiment (E)

A procedure performed that produces a result.

Example: Tossing a coin five times (E_5). L4

Definition: Outcome (ξ)

A possible result of an experiment.

Example: One outcome of E_5:
ξ_1 = HHTHT L4

Definition: Event

A set of outcomes of an experiment.

Example: Event C in experiment E_5
C = {all outcomes consisting of an even number of heads} L4

3.2 Sample Space

Definition: Sample Space (S)

The collection (set) of all possible distinct outcomes of an experiment. L4

Properties:

- Outcomes are **mutually exclusive**
  → Only one outcome occurs at a time.
  Example: coin flip → heads OR tails.
- Outcomes are **collectively exhaustive**
  → No other outcomes are possible. L4

Additional Notes on Sample Space

- S is the universal set of outcomes.
- Sample space may be:
  - Discrete
  - Countably infinite
  - Continuous L4

Examples of Sample Spaces L4

1. Flipping a fair coin once
2. Rolling a cubical die
3. Rolling two dice
4. Flipping a coin until a tail occurs
5. Random number generator in interval [0,1]

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4. Probability and Axioms

4.1 Probability

Probability measures the likelihood of events.

It is:

- a measure of likelihood, or
- a function assigning a numerical value to an event's likelihood. [L4]

4.2 Axioms of Probability

(Accepted without proof)

For any event A:

Axiom 1: Bounds

$$0 \leq Pr(A) \leq 1$$

Axiom 2: Certainty

If S is the sample space,

$$Pr(S) = 1$$

Axiom 3: Additivity

If $A \cap B = \emptyset$,

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For infinitely many mutually exclusive events A_i:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

[L4]

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5. Corollary from Probability Axioms

Define $A_i = \emptyset$ for all $i > n$.

Corollary 2.1

For a finite number (M) of mutually exclusive events:

$$Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} Pr(A_i)$$

[L4]

Notes:

- Equivalent to Axiom 3 when the sample space is finite.

- Axiom 3's infinite form is needed for infinite sample spaces. L4

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6. Propositions from Probability Axioms

Proposition 2.1 (Complement Rule)

$$Pr(A^c) = 1 - Pr(A)$$

L4

Proposition 2.2 (Monotonicity)

If $A \subset B$, then:

$$Pr(A) \leq Pr(B)$$

L4

✔️ Key Takeaways (Exam Focus)

- Experiment → procedure producing results.
- Outcome → single result.
- Event → set of outcomes.
- Sample space → all possible outcomes.
- Outcomes must be mutually exclusive & collectively exhaustive.
- Probability satisfies three axioms.
- Complement and subset relations give useful probability properties.

If you want, I can next:

✅ simplify this for **quick revision**
✅ create **important exam questions** from it
✅ explain joint & conditional probability when you upload the next slides
✅ convert it into **LaTeX format for submission**

Sources

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CSE400 – Fundamentals of Probability in Computing

Lecture 4: Joint Probability and Conditional Probability L4

Instructor: Dhaval Patel, PhD
Date: January 15, 2026 L4

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1. Course Context and Motivation

1.1 Why Learn CSE400?

- Daily life conversations involve uncertainty. L4

1.2 Engineering Applications L4

- Speech Recognition
- System Radar System
- Communication Network

Speech Recognition System

Speech templates for words (Hello, Yes, No, Bye) are compared to recognize spoken input across different speakers. L4

Radar System

Radar systems detect objects by transmitting signals and analyzing reflections. L4

Communication Network

Information travels from a **source** to a **destination** through interconnected nodes. L4

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2. Lecture Outline L4

Introduction to Probability Theory

- Experiments, Sample Space, and Events
- Axioms of Probability
- Corollaries and Propositions from Probability Axioms
- How to Assign Probability: Classical and Relative Frequency Approaches

Joint Probability

- Motivation, Notation, and Concepts of Joint Probability
- Example 1: Card Deck Example
- Example 2: Costume Party Example

Conditional Probability

- Motivation, Notations, and Concepts of Conditional Probability
- Example 3: Cards Without Replacement
- Example 4: Game of Poker
- Example 5: The Missing Key

(Slides provided cover the Introduction to Probability Theory section.)

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3. Introduction to Probability Theory

3.1 Experiments, Outcomes, and Events

Definition: Experiment (E)

A procedure we perform that produces some result.

Example: Tossing a coin five times (E_5).
L4

Definition: Outcome (ξ)

A possible result of an experiment.

Example: One outcome of E_5 is
ξ_1 = HHTHT
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Definition: Event

A certain set of outcomes of an experiment.

Example: Event C with experiment E_5
C = {all outcomes consisting of an even number of heads}
L4

3.2 Sample Space

Definition: Sample Space (S)

The collection or set of all possible distinct outcomes of an experiment. L4

Properties:

- **Mutually Exclusive:** You can get heads or tails, but not both.
- **Collectively Exhaustive:** You cannot get anything other than heads or tails. L4

Additional Notes

- S is the universal set of outcomes.
- Sample space may be:
  - Discrete
  - Countably infinite
  - Continuous L4

Examples of Sample Spaces L4

1. Flipping a fair coin once
2. Cubical die with numbered faces is rolled
3. Rolling two dices
4. Flip a coin until a tails occurs
5. Random number generator with interval [0, 1]

---

4. Probability and Axioms

4.1 Probability

Probability is a measure of the likelihood of various events OR a function of an event that produces a numerical quantity that measures the likelihood of that event. L4

There are many ways to define such a function.

4.2 Axioms of Probability

The statements are taken to be self-evident and require no proof. L4

Axiom 1 (Bounds)

For any event A:

$$0 \leq Pr(A) \leq 1$$

Axiom 2 (Certainty)

If S is the sample space for a given experiment:

$$Pr(S) = 1$$

Axiom 3 (Additivity)

If $A \cap B = \emptyset$:

$$Pr(A \cup B) = Pr(A) + Pr(B)$$

For infinitely many mutually exclusive sets A_i, $i = 1, 2, 3, \ldots$, with $A_i \cap A_j = \emptyset$ for all $i \neq j$:

$$Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} Pr(A_i)$$

L4

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5. Corollary from Probability Axioms L4

Define A_i as the null event for all values of i greater than n.

Corollary 2.1

For **M finite number of mutually exclusive sets** A_i, $i = 1, 2, 3, \ldots, M$, with $A_i \cap A_j = \emptyset$ for all $i \neq j$:

$$Pr\left(\bigcup_{i=1}^{M} A_i\right) = \sum_{i=1}^{M} Pr(A_i)$$

Notes:

- Axiom 3 is equivalent to Corollary 2.1 when the sample space is finite.
- The added generality of Axiom 3 is necessary when the sample space consists of an infinite number of points.

---

6. Propositions from Probability Axioms

Proposition 2.1 (Complement Rule)

$$Pr(A^c) = 1 - Pr(A)$$

L4

Proposition 2.2 (Monotonicity)

If $A \subset B$, then:

$$Pr(A) \leq Pr(B)$$

L4

✔️ Summary for Exam Preparation

- Experiment → procedure producing a result.
- Outcome → possible result.
- Event → set of outcomes.
- Sample space → all possible outcomes.
- Outcomes must be mutually exclusive and collectively exhaustive.
- Probability satisfies three axioms.
- Complement rule and subset relation follow from axioms.
- Corollary simplifies additivity for finite cases.

If you want, I can next:

✅ convert this into **LaTeX scribe format**
✅ create **exam questions & answers**
✅ explain joint & conditional probability when the next slides are provided
✅ give a **super short revision sheet**

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- Example hookup: Costume party

Conditional Probability

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✔️ Key Points for Exam Preparation

- Experiment → procedure producing results.
- Outcome → possible result.
- Event → set of outcomes.
- Sample space → all possible outcomes.
- Outcomes must be mutually exclusive and collectively exhaustive.
- Probability obeys three axioms.
- Additivity applies to mutually exclusive events.
- Complement and subset properties follow from axioms.
- Finite additivity is given by Corollary 2.1.

If you want, I can next:

- convert this into **LaTeX lecture scribe format**
- produce a **one-page revision sheet**
- generate **likely exam questions**
- explain joint & conditional probability when you upload the next slides

Sources